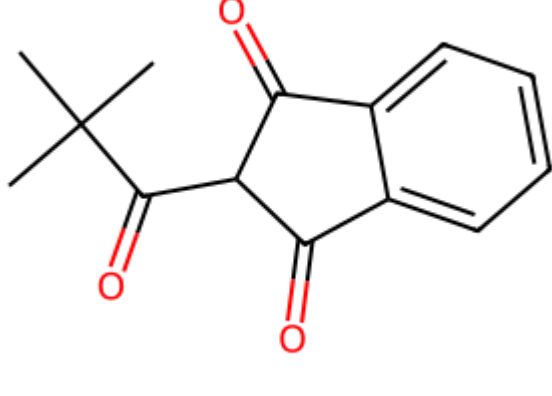
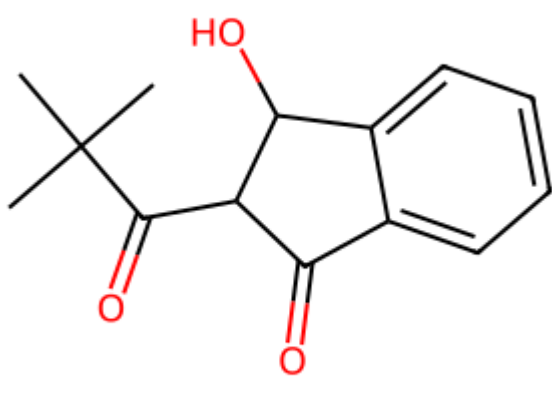




```
{'analysis_time': 0.0,  
'first_solution_time': 0.192378,  
'num_calls_rxn_model': 0,  
'num_calls_value_function': 0,  
'reaction_number_estimate': 0.0,  
'retro_star_min_cost': 3.2874223674033107,  
'retro_star_mol_cost': inf,  
'retro_star_reaction_number': 3.2874223674033107,  
'retro_star_value': 3.2874223674033107}
```

Reaction

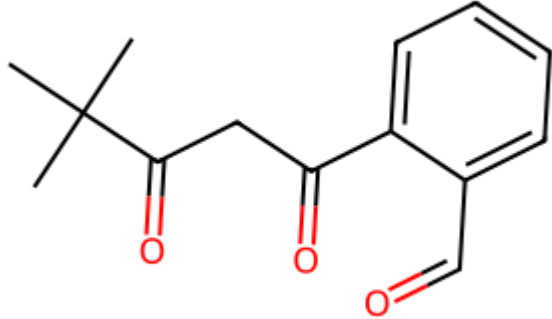
```
{'analysis_time': 0.192378,  
'first_solution_time': 9.943251,  
'num_calls_rxn_model': 1,  
'num_calls_value_function': 0,  
'retro_star_min_cost': 7.646913761926838,  
'retro_star_reaction_number': 7.646913761926838,  
'retro_star_rxn_cost': 3.3813947543659753,  
'retro_star_value': 7.646913761926838}
```



```
{'analysis_time': 0.192378,  
'first_solution_time': 9.943251,  
'num_calls_rxn_model': 1,  
'num_calls_value_function': 0,  
'reaction_number_estimate': 0.0,  
'retro_star_min_cost': 4.265519007560862,  
'retro_star_mol_cost': inf,  
'retro_star_reaction_number': 4.265519007560862,  
'retro_star_value': 7.646913761926838}
```

Reaction

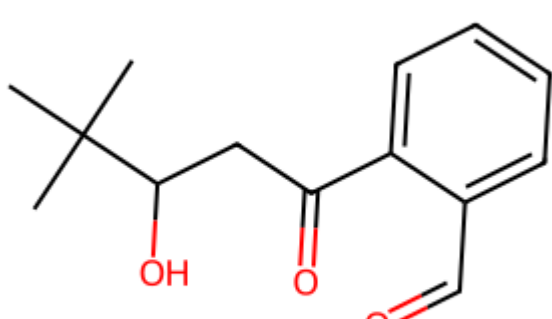
```
{'analysis_time': 2.14598,  
'first_solution_time': 9.943251,  
'num_calls_rxn_model': 13,  
'num_calls_value_function': 332,  
'retro_star_min_cost': 4.265519007560862,  
'retro_star_reaction_number': 4.265519007560862,  
'retro_star_rxn_cost': 0.9915532163747018,  
'retro_star_value': 7.646913761926838}
```



```
{'analysis_time': 2.14598,  
'first_solution_time': 9.943251,  
'num_calls_rxn_model': 13,  
'num_calls_value_function': 332,  
'reaction_number_estimate': 0.0,  
'retro_star_min_cost': 3.2739657911861606,  
'retro_star_mol_cost': inf,  
'retro_star_reaction_number': 3.2739657911861606,  
'retro_star_value': 7.646913761926838}
```

Reaction

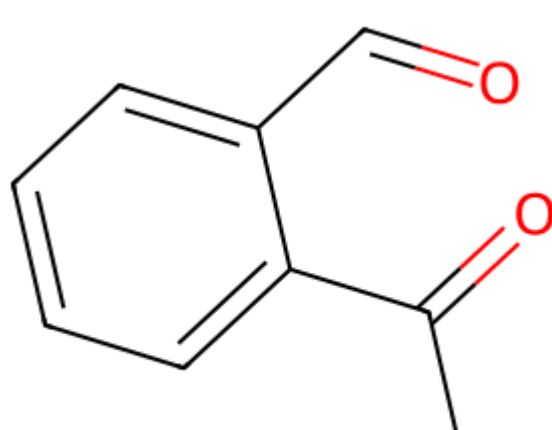
```
{'analysis_time': 2.314832,  
'first_solution_time': 9.943251,  
'num_calls_rxn_model': 14,  
'num_calls_value_function': 364,  
'retro_star_min_cost': 3.2739657911861606,  
'retro_star_reaction_number': 3.2739657911861606,  
'retro_star_rxn_cost': 2.2633643798407643,  
'retro_star_value': 7.646913761926838}
```



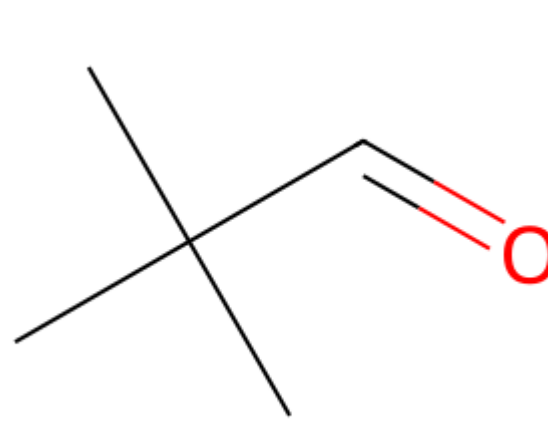
```
{'analysis_time': 2.314832,  
'first_solution_time': 9.943251,  
'num_calls_rxn_model': 14,  
'num_calls_value_function': 364,  
'reaction_number_estimate': 0.0,  
'retro_star_min_cost': 1.0106014113453963,  
'retro_star_mol_cost': inf,  
'retro_star_reaction_number': 1.0106014113453963,  
'retro_star_value': 7.646913761926838}
```

Reaction

```
{'analysis_time': 9.943251,  
'first_solution_time': 9.943251,  
'num_calls_rxn_model': 61,  
'num_calls_value_function': 1525,  
'retro_star_min_cost': 1.0106014113453963,  
'retro_star_reaction_number': 1.0106014113453963,  
'retro_star_rxn_cost': 1.0106014113453963,  
'retro_star_value': 7.646913761926838}
```



```
{'analysis_time': 9.943251,  
'first_solution_time': 9.943251,  
'num_calls_rxn_model': 61,  
'num_calls_value_function': 1525,  
'retro_star_min_cost': 0.0,  
'retro_star_mol_cost': 0.0,  
'retro_star_reaction_number': 0.0,  
'retro_star_value': 7.646913761926838}
```



```
{'analysis_time': 9.943251,  
'first_solution_time': 9.943251,  
'num_calls_rxn_model': 61,  
'num_calls_value_function': 1525,  
'retro_star_min_cost': 0.0,  
'retro_star_mol_cost': 0.0,  
'retro_star_reaction_number': 0.0,  
'retro_star_value': 7.646913761926838}
```